

Proposition-A evidence-chain migration and re-validation record (migration batch 1)

TECT verification-first repository · claim B2-PROPA-HLAYER

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1. Purpose and scope

This note is the permanent record of migration batch 1 (plan phase M1, [governance/migration-plan.md](#) §4 priority 1): the evidence chain of claims **A1-KERNEL-CONV**, **B1-RH-ENUM** (partial), and **B2-PROPA-HLAYER** was copied verbatim from the frozen legacy repository into `archive/legacy/` and re-validated by full re-execution. This note makes no new mathematical claim and performs no tier action; it is citable as the re-validation evidence for the cards above.

2. What was migrated

23 files (11 notes, 7 scripts, 5 run artefacts) covering theory tags Math426 (+AddA/AddB), Math435 (v1.0–v1.1), Math437 (v1.0–v1.2), Math440, Math441, Math442, plus the three import-dependency modules Math374 / Math400-AddE / Math424-AddA (script-only). Layout: `archive/legacy/{notes/<Tag>/, scripts/, artefacts/<Tag>/}`; per-file original paths in `archive/MIGRATION-LEDGER.md`; per-tag lookup in `archive/legacy/INDEX.md`.

3. Re-validation results

Fresh environment: python 3.12, numpy 2.2.6.

Script	Asserts	Runtime	Artefact comparison
Math426_g4_kernel_reconciliation.py	10/10	0.4 s	identical, rel. tol. 10^{-9}
Math435_g6_corrected_variable_cascade.py	101/101	3.5 s	identical, rel. tol. 10^{-9}
Math437_step5_class_closure.py	91/91	1.2 s	numerics identical; 2 verdict-string diffs (§4)
Math440_audit_secondwave_recheck.py	75/75	37.8 s	identical, rel. tol. 10^{-9}

Total: **277/277 asserts PASS**. Fresh artefacts: `claims/A1-KERNEL-CONV/runs/260605-migration-revalidation/`, `claims/B2-PROPA-HLAYER/runs/260605-migration-revalidation/`. After the archive per-tag reorganisation, `Math426_g4_kernel_reconciliation.py` was re-executed from the new `archive/legacy/scripts/` location (10/10 PASS), confirming the flat script directory remains runnable.

4. Finding F-1 (STALE-ARTEFACT, no numeric impact)

The archived legacy artefact [artefacts/Math437/step5_class_closure.json](#) predates the R1 repair of the Math440/Math441 audit cycle: its `claims[90]` verdict string is the v1.0-era wording (“Region-I positivity + Lemma-B dip bound + Δ_0 penalty”), while the migrated script (repaired, v1.1 four-interval proof) emits “I’ all-positivity + band/Region-II anchor-only P_B floors; Δ_0 demoted to comfort margin”. All numeric fields are identical within 10^{-9} . Consequence: the fresh artefact under [claims/B2-PROPA-HLAYER/runs/260605-migration-revalidation/](#) is canonical for TSv2 citation; the archived JSON remains as historical record. Process lesson: migration batches MUST re-run scripts rather than trust stored artefacts (binding via [governance/migration-plan.md](#) §3, M1).

5. Hypothesis transcription

H-LAYER and H-A0 were transcribed verbatim from Math437 v1.2 §Hypotheses into [claims/GATES.md](#). The transcription makes explicit that the H-LAYER beyond-layer residual *is* gate STEP-5B — the designated gateway for any whole-Reading-H T6 discussion.

6. Devil’s-advocate (three objections)

- (α) “*Re-execution in the same ecosystem is not independent verification.*” VALID with mitigation: the asserts are the original authors’ self-tests, so this re-run certifies reproducibility, not correctness. Independent correctness evidence for the chain is the dual-audit record (Math440 + Math441, different agents) migrated alongside. External reproduction becomes possible exactly because of this migration; CI re-execution on the GitHub runner adds an environment-independent check.
- (β) “*One stale artefact suggests others.*” VALID: treated as a class, not an instance — the re-run-don’t-trust rule is binding for every future batch, and any artefact cited without a fresh re-run is non-citable ([governance/verification-standard.md](#) §4).
- (γ) “*Archive copies could drift from the frozen legacy originals.*” DISMISSED: the legacy repository is frozen read-only (M0); copies are byte-identical at migration time, recorded in the ledger, and git-tracked from the first public commit — any later divergence is visible in history.

Quantitative sanity check (mandatory): the four assert totals ($10 + 101 + 91 + 75 = 277$) and the 10^{-9} comparison tolerance were checked against the comparison-script output; runtimes are $O(1\text{--}40\text{s})$, consistent with the scripts’ loop sizes.

7. Result footer

Result ID:	B2-PROPA-HLAYER (supporting record; also A1-KERNEL-CONV, B1-RH-ENUM partial)
Precise statement:	Migration batch-1 evidence chain migrated verbatim and re-validated by full re-execution (277/277).
Scope:	Reproducibility of the migrated chain only; no new mathematical claim; no tier action.
Dependencies:	archive/legacy/ batch-1 file set (ledger 2026-06-05).

Evidence grade:	EXECUTED
Reproduction command:	cd archive/legacy/scripts && python Math426_g4_kernel_reconciliation.py && python Math435_g6_corrected_variable_cascade.py && python Math437_step5_class_closure.py && python Math440_audit_secondwave_recheck.py
Expected output:	10/10, 101/101, 91/91, 75/75 asserts PASS; exit 0; JSONs match archived artefacts within rel_tol 1e-9 (Math437: numerics only, see Section 4)
Falsification gate:	Any assert failure or artefact mismatch beyond tolerance on re-execution.
Tier before / after:	A1 T5 / T5 ; B1 T5 / T5 ; B2 T6 / T6 (no change)
No-overclaim statement:	Re-execution certifies reproducibility, not correctness; Reading-H remains T5; STEP-5B open.
Next required action:	Migration batch 2 (Math431-HEX chain + Math427/428-432/434/436) for B1 cleanness.